



Top 34 CAT Scheduling Questions With Video Solutions

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Questions

Instructions

A visa processing office (VPO) accepts visa applications in four categories - US, UK, Schengen, and Others. The applications are scheduled for processing in twenty 15- minute slots starting at 9:00 am and ending at 2:00 pm. Ten applications are scheduled in each slot.

There are ten counters in the office, four dedicated to US applications, and two each for UK applications, Schengen applications and Others applications. Applicants are called in for processing sequentially on a first-come-first-served basis whenever a counter gets freed for their category. The processing time for an application is the same within each category. But it may vary across the categories. Each US and UK application requires 10 minutes of processing time. Depending on the number of applications in a category and time required to process an application for that category, it is possible that an applicant for a slot may be processed later.

On a particular day, Ira, Vijay and Nandini were scheduled for Schengen visa processing in that order. They had a 9:15 am slot but entered the VPO at 9:20 am. When they entered the office, exactly six out of the ten counters were either processing applications, or had finished processing one and ready to start processing the next.

Mahira and Osman were scheduled in the 9:30 am slot on that day for visa processing in the Others category.

The following additional information is known about that day.

1. All slots were full.
2. The number of US applications was the same in all the slots. The same was true for the other three categories.
3. 50% of the applications were US applications.
4. All applicants except Ira, Vijay and Nandini arrived on time.
5. Vijay was called to a counter at 9:25 am.

Question 1

What is the maximum possible value of the total time (in minutes, nearest to its integer value) required to process all applications in the Others category on that day?

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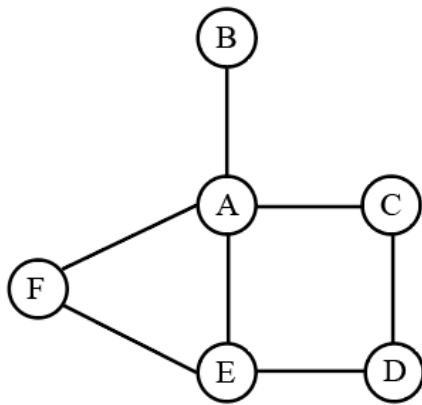
1. All slots were full.
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3. 50% of the applications were US applications.

VIDEO SOLUTION

Instructions

Comprehension:

A, B, C, D, E and F are the six police stations in an area, which are connected by streets as shown below. Four teams - Team 1, Team 2, Team 3 and Team 4 - patrol these streets continuously between 09:00 hrs. and 12:00 hrs. each day.



The teams need 30 minutes to cross a street connecting one police station to another. All four teams start from Station A at 09:00 hrs. and must return to Station A by 12:00 hrs. They can also pass via Station A at any point on their journeys.

The following facts are known.

1. None of the streets has more than one team traveling along it in any direction at any point in time.
2. Teams 2 and 3 are the only ones in stations E and D respectively at 10:00 hrs.
3. Teams 1 and 3 are the only ones in station E at 10:30 hrs.
4. Teams 1 and 4 are the only ones in stations B and E respectively at 11:30 hrs.
5. Team 1 and Team 4 are the only teams that patrol the street connecting stations A and E.
6. Team 4 never passes through Stations B, D or F.

Question 2

How many times do the teams pass through Station B in a day?

	9	9:30	10	10:30	11	11:30	12
Team 1	A						A
Team 2	A		E				A
Team 3	A		D				A
Team 4	A						A

VIDEO SOLUTION

Instructions

The Humanities department of a college is planning to organize eight seminars, one for each of the eight doctoral students - A, B, C, D, E, F, G and H. Four of them are from Economics, three from Sociology and one from Anthropology department. Each student is guided by one among P, Q, R, S and T. Two students are guided by each of P, R and T, while one student is guided by each of Q and S. Each student is guided by a guide belonging to their department. Each seminar is to be scheduled in one of four consecutive 30-minute slots starting at 9:00 am, 9:30 am, 10:00 am and 10:30 am on the same day. More than one seminars can be scheduled in a slot, provided the guide is free. Only three rooms are available and hence at the most three seminars can be scheduled in a slot. Students who are guided by the same guide must be scheduled in consecutive slots.

The following additional facts are also known.

1. Seminars by students from Economics are scheduled in each of the four slots.
2. A's is the only seminar that is scheduled at 10:00 am. A is guided by R.
3. F is an Anthropology student whose seminar is scheduled at 10:30 am.
4. The seminar of a Sociology student is scheduled at 9:00 am.

5. B and G are both Sociology students, whose seminars are scheduled in the same slot. The seminar of an Economics student, who is guided by T, is also scheduled in the same slot.
6. P, who is guiding both B and C, has students scheduled in the first two slots.
7. A and G are scheduled in two consecutive slots.

Question 3

If D is scheduled in the slot immediately before Q's, then which of the following is NOT necessarily true?

- A G is guided by Q.
- B E is guided by R.
- C F is guided by S.
- D D is guided by T.



Handwritten notes on the left:

- (2) P -
- (1) Q -
- (2) R -
- (1) S -
- (2) T -

Printed text on the right:

10:30 am on the same day. More than one seminars can be scheduled in a slot, provided the guide is free. Only three rooms are available and hence at the most three seminars can be scheduled in a slot. Students who are guided by the same guide must be scheduled in consecutive slots.

The following additional facts are also known.

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- A's seminar is scheduled at 10:00 am. A is guided by R.
- F is an Anthropology student whose seminar is scheduled at 10:30 am.
- The seminar of a Sociology student is scheduled at 9:00 am.
- B and G are both Sociology students, whose seminars are scheduled in the same slot. The seminar of an Economics student, who is guided by T, is scheduled in the same slot.
- P, who is guiding both B and C, has students scheduled in the first two slots.
- A and G are scheduled in two consecutive slots.

Table:

9:00	
9:30	
10:00	
10:30	



VIDEO SOLUTION



CAT 2024 STUDY PLAN

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Question 4

Who all are NOT guiding any Economics students?

- A Q, R and S
- B P, Q and R
- C P, R and S
- D P, Q and S

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(2) P -
 (1) Q -
 (2) R -
 (1) S -
 (2) T -

9:00	
9:30	
10:00	
10:30	

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 6. The seminar of a student guided by both R and C, has students scheduled in the first two slots.
 7. B and G are scheduled in two consecutive slots.

VIDEO SOLUTION

Question 5

Which one of the following statements is true?

- A Two seminars are scheduled in the first slot.
- B Only one seminar is scheduled in the second slot.
- C Three seminars are scheduled in the first slot.
- D Three seminars are scheduled in the last slot.

Cracku

(2) P -
 (1) Q -
 (2) R -
 (1) S -
 (2) T -

9:00	
9:30	
10:00	
10:30	

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Mahira and Osman were scheduled in the 9:30 am slot on that day for visa processing in the Others category.

The following additional information is known about that day.

1. All slots were full.
2. The number of US applications was the same in all the slots. The same was true for the other three categories.
3. 50% of the applications were US applications.
4. All applicants except Ira, Vijay and Nandini arrived on time.
5. Vijay was called to a counter at 9:25 am.

Question 6

When did the application processing for all US applicants get over on that day?

- A 2:05 pm
- B 2:25 pm
- C 2:00 pm
- D 3:40 pm

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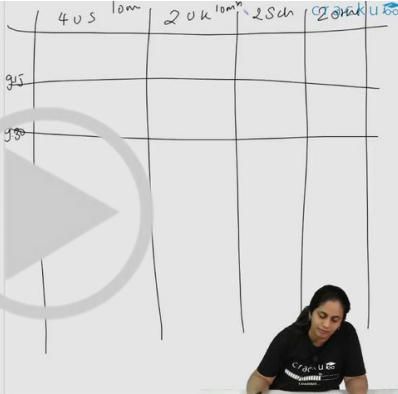
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VIDEO SOLUTION

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Question 7

How many UK applications were scheduled on that day?

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VIDEO SOLUTION

Question 8

Which of the following statements is false?

- A The application process of Osman was completed before 9:45 am.
- B The application process of Mahira started after Nandini's.
- C The application process of Osman was completed before Vijay's.
- D The application process of Mahira was completed before Nandini's.

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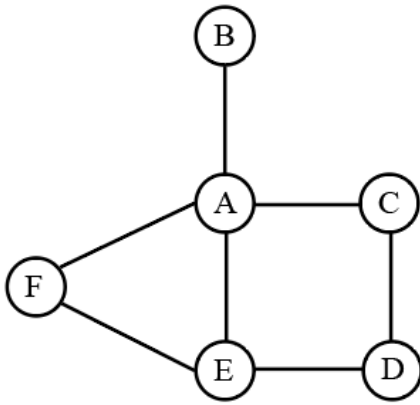
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The teams need 30 minutes to cross a street connecting one police station to another. All four teams start from Station A at 09:00 hrs. and must return to Station A by 12:00 hrs. They can also pass via Station A at any point on their journeys.

The following facts are known.

1. None of the streets has more than one team traveling along it in any direction at any point in time.
2. Teams 2 and 3 are the only ones in stations E and D respectively at 10:00 hrs.
3. Teams 1 and 3 are the only ones in station E at 10:30 hrs.
4. Teams 1 and 4 are the only ones in stations B and E respectively at 11:30 hrs.
5. Team 1 and Team 4 are the only teams that patrol the street connecting stations A and E.
6. Team 4 never passes through Stations B, D or F.

Question 9

How many times does Team 4 pass through Station E in a day?

	9	9:30	10	10:30	11	11:30	12
Team 1	A						A
Team 2	A		E				A
Team 3	A		D				A
Team 4	A						A

VIDEO SOLUTION

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Question 10

How many teams pass through Station C in a day?

- A 4
- B 3
- C 1
- D 2

VIDEO SOLUTION

Instructions

Healthy Bites is a fastfood joint serving three items, burgers, fries and ice cream. It has two employees Anish and Bani who prepare the items ordered by the clients. Preparation time is 10 minutes for a burger and 2 minutes for an order of Ice cream. An employee can prepare only one of these items at a time. The fries are prepared in an automatic fryer which can prepare upto to 3 portions of fries at a time, and takes 5 minutes irrespective of the number of portions. The fryer does not need an employee to constantly attend to it, and we can ignore the time taken by an employee to start and stop the fryer; thus, an employee can be engaged in preparing other items while the frying is on. However fries cannot be prepared in anticipation of future orders.

Healthy Bites wishes to serve the orders as early as possible. The individual items in any order are served as and when ready; however, the order is considered to be completely served only when all the items of that order are served.

The table below gives the orders of three clients and the times at which they placed their orders:

Client no.	Time	Order
1	10:00	1 burger, 3 portions of fries, 1 order of ice cream
2	10:05	2 portions of fries. 1 order of ice cream
3	10:07	1 burger. 1 portion of fries

Question 11

Assume that only one client's order can be processed at any given point of time. So, Anish or Bani cannot start preparing a new order while a previous order is being prepared.

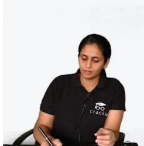
At what time is the order placed by Client 1 completely served?

- A 10:17
- B 10:10
- C 10:15
- D 10:20

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Assume that only one client's order can be processed at any given point of time. So, Anish or Bani cannot start preparing a new order while a previous order is being prepared.
At what time is the order placed by Client 1 completely served?

+

VIDEO SOLUTION

Question 12

Suppose the employees are allowed to process multiple orders at a time, but the preference would be to finish orders of clients who placed their orders earlier.

Also assume that the fourth client came in only at 10:35. Between 10:00 and 10:30, for how many minutes is exactly one of the employees idle?

- A 7
- B 10
- C 15
- D 23

cracku

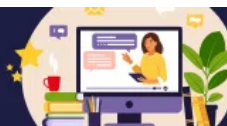
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+




VIDEO SOLUTION

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Instructions

A journal plans to publish 18 research papers, written by eight authors (A, B, C, D, E, F, G, and H) in four issues of the journal scheduled in January, April, July and October. Each of the research papers was written by exactly one of the eight authors. Five papers were scheduled in each of the first two issues, while four were scheduled in each of the last two issues. Every author wrote at least one paper and at most three papers. The total number of papers written by A, D, G and H was double the total number of papers written by the other four authors. Four

of the authors were from India and two each were from Japan and China. Each author belonged to exactly one of the three areas – Manufacturing, Automation, and Logistics. Four of the authors were from the Logistics area and two were from the Automation area. As per the journal policy, none of the authors could have more than one paper in any issue of the journal.

The following facts are also known.

1. F, an Indian author from the Logistics area, wrote only one paper. It was scheduled in the October issue.
2. A was from the Automation area and did not have a paper scheduled in the October issue.
3. None of the Indian authors were from the Manufacturing area and none of the Japanese or Chinese authors were from the Automation area.
4. A and H were from different countries, but had their papers scheduled in exactly the same issues.
5. C and E, both Chinese authors from different areas, had the same number of papers scheduled. Further, E had papers scheduled in consecutive issues of the journal but C did not.
6. B, from the Logistics area, had a paper scheduled in the April issue of the journal.
7. B and G belonged to the same country. None of their papers were scheduled in the same issue of the journal.
8. D, a Japanese author from the Manufacturing area, did not have a paper scheduled in the July issue.
9. C and H belonged to different areas.

Question 13

What is the correct sequence of number of papers written by B, C, E and G, respectively?

- A 1, 2, 2, 3
- B 1, 3, 3, 1
- C 3, 1, 1, 3
- D 1, 2, 2, 1

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$A+D+G+H = 2(B+C+E+F)$

$2x + 11 = 2(2x + 18)$

$2x + 11 = 4x + 36$

$-2x = 25$

$x = -12.5$

Jan - 5
April - 5
July - 4
Oct - 4

$x + 2x = 218$

226

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8. D, a Japanese author from the Manufacturing area, did not have a paper scheduled in the July issue.
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VIDEO SOLUTION

Question 14

Which of the following statement(s) MUST be true?

Statement A: Every issue had at least one paper by author(s) from each country.

Statement B: Every issue had at most two papers by author(s) from each area.

- A Both the statements
- B Only Statement B
- C Only Statement A
- D Neither of the statements

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$$A + D + G + H = 2(B + E + F)$$

Jan - 5
April - 5
July - 4
Oct - 4
Total = 18

$x + 2x = 18$
 $3x = 18$
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VIDEO SOLUTION

Question 15

Which of the following statements is FALSE?

- A Every issue had at least one paper by author(s) from Automation area.
- B Every issue had exactly one paper by a Chinese author.
- C Every issue had exactly two papers by authors from Logistics area.
- D Every issue had exactly two papers by Indian authors.

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8. F, from the Logistics area, had a paper scheduled in the January issue of the journal.

VIDEO SOLUTION

CAT 2024 Coaching By 4 Time CAT 100%iler & IIM Alumni

Question 16

Which of the following is the correct sequence of number of papers by authors from Automation, Manufacturing and Logistics areas, respectively?

- A 6, 5, 7
- B 6, 6, 6
- C 6, 7, 5
- D 5, 6, 7

cracku

$A+D+G+H = 2(B+E+F)$

$2x = 12$

$x = 6$

Jan - 5

April - 5

July - 4

Oct - 4

$x + 2x = 18$

$2 > 6$

A journal plans to publish 18 research papers, written by eight authors (A, B, C, D, E, F, G, and H) in four issues of the journal scheduled in January, April, July and October. Each of the research papers was written by exactly one of the eight authors. Five papers were scheduled in each of the first two issues, while four were scheduled in each of the last two issues. Every author wrote at least one paper and at most three papers. The total number of papers written by A, D, G and H was double the total number of papers written by the other four authors. Four of the authors were from India and two were from Japan and China. Each author belonged to exactly one of the three areas: Manufacturing, Automation, and Logistics. Four of the authors were from the Logistics area and two were from the Automation area. As per the journal policy, none of the authors could have more than one paper in any issue of the journal.

1. The authors are also known. The author from the Logistics area, wrote only one paper. It was scheduled in the first issue.

2. The author from the Automation area and did not have a paper scheduled in the first issue.

3. None of the Indian authors were from the Manufacturing area and none of the Japanese or Chinese authors were from the Automation area.

4. A and H were from different countries, but had their papers scheduled in exactly the same issue.

5. C and E, both Chinese authors from different areas, had the same number of papers scheduled. Further, E had papers scheduled in consecutive issues of the journal but C did not.

6. B, from the Logistics area, had a paper scheduled in the April issue of the journal.

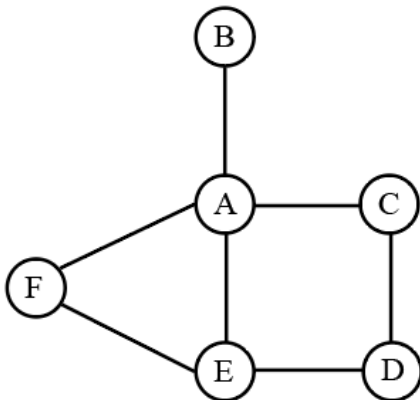
7. B and G belonged to the same country. None of their papers were scheduled in the same issue of the journal.

VIDEO SOLUTION

Instructions

Comprehension:

A, B, C, D, E and F are the six police stations in an area, which are connected by streets as shown below. Four teams - Team 1, Team 2, Team 3 and Team 4 - patrol these streets continuously between 09:00 hrs. and 12:00 hrs. each day.



The teams need 30 minutes to cross a street connecting one police station to another. All four teams start from Station A at 09:00 hrs. and must return to Station A by 12:00 hrs. They can also pass via Station A at any point on their journeys.

The following facts are known.

1. None of the streets has more than one team traveling along it in any direction at any point in time.
2. Teams 2 and 3 are the only ones in stations E and D respectively at 10:00 hrs.
3. Teams 1 and 3 are the only ones in station E at 10:30 hrs.
4. Teams 1 and 4 are the only ones in stations B and E respectively at 11:30 hrs.
5. Team 1 and Team 4 are the only teams that patrol the street connecting stations A and E.
6. Team 4 never passes through Stations B, D or F.

Question 17

Which team patrols the street connecting Stations D and E at 10:15 hrs?

- A Team 4
- B Team 1
- C Team 2
- D Team 3

	9	9:30	10	10:30	11	11:30	12
T1	A		E				A
T2	A		E				A
T3	A		D				A
T4	A						A

The teams need to cross a street connecting one police station to another. All four teams start at Station A at 09:00 hrs. and must return to Station A by 12:30 hrs. They can only travel in one direction on their journeys.

Additional facts are known:

- There is only one team travelling along it in any direction at any time.
- Teams 1 and 2 are the only teams that pass through Station E at 10:00 hrs.
- Teams 3 and 4 are the only teams that pass through Station E at 10:30 hrs.
- Teams 1 and 2 are the only teams that pass through Station B at 11:30 hrs.
- Teams 3 and 4 are the only teams that pass through Station D at 11:30 hrs.
- Team 4 never passes through Stations B, D or F.

VIDEO SOLUTION

Instructions

A visa processing office (VPO) accepts visa applications in four categories - US, UK, Schengen, and Others. The applications are scheduled for processing in twenty 15- minute slots starting at 9:00 am and ending at 2:00 pm. Ten applications are scheduled in each slot.

There are ten counters in the office, four dedicated to US applications, and two each for UK applications, Schengen applications and Others applications. Applicants are called in for processing sequentially on a first-come-first-served basis whenever a counter gets freed for their category. The processing time for an application is the same within each category. But it may vary across the categories. Each US and UK application requires 10 minutes of processing time. Depending on the number of applications in a category and time required to process an application for that category, it is possible that an applicant for a slot may be processed later.

On a particular day, Ira, Vijay and Nandini were scheduled for Schengen visa processing in that order. They had a 9:15 am slot but entered the VPO at 9:20 am. When they entered the office, exactly six out of the ten counters were either processing applications, or had finished processing one and ready to start processing the next.

Mahira and Osman were scheduled in the 9:30 am slot on that day for visa processing in the Others category.

The following additional information is known about that day.

1. All slots were full.
2. The number of US applications was the same in all the slots. The same was true for the other three categories.
3. 50% of the applications were US applications.
4. All applicants except Ira, Vijay and Nandini arrived on time.
5. Vijay was called to a counter at 9:25 am.

Question 18

Which of the following is the closest to the time when Nandini's application process got over?

- A 9:50 am
- B 9:37 am
- C 9:35 am
- D 9:45 am

A visa processing office (VPO) accepts visa applications in four categories - US, UK, Schengen, and Others. The applications are scheduled for processing in twenty 15-minute slots starting at 9:00 am and ending at 2:00 pm. Ten applications are scheduled in each slot.

There are ten counters in the office, four dedicated to US applications, and two each for UK applications, Schengen applications and Others applications. Applicants are called in for processing sequentially on a first-come-first-served basis whenever a counter gets freed for their category. The processing time for an application is the same within each category. But it may vary across the categories. Each US application requires 10 minutes of processing time. Depending on the number of applications in a category and time required to process an application for that category, it is possible that an applicant's slot may be processed later.

On a particular day, Ira, Vijay and Nandini were scheduled for Schengen visa processing in that order. They had a 9:15 am slot but entered the VPO at 9:20 am. When they entered the office, exactly six out of the ten counters were either processing applications, or had finished processing one and ready to start processing the next.

Mahira and Osman were scheduled in the 9:30 am slot on that day for visa processing in the Others category.

The following additional information is known about that day.

1. All slots were full.
2. The number of US applications was the same in all the slots. The same was true for the other three categories.
3. 50% of the applications were US applications.

VIDEO SOLUTION

CAT Previous Papers PDF

Instructions

Healthy Bites is a fastfood joint serving three items, burgers, fries and ice cream. It has two employees Anish and Bani who prepare the items ordered by the clients. Preparation time is 10 minutes for a burger and 2 minutes for an order of Ice cream. An employee can prepare only one of these items at a time. The fries are prepared in an automatic fryer which can prepare upto to 3 portions of fries at a time, and takes 5 minutes irrespective of the number of portions. The fryer does not need an employee to constantly attend to it, and we can ignore the time taken by an employee to start and stop the fryer; thus, an employee can be engaged in preparing other items while the frying is on. However fries cannot be prepared in anticipation of future orders.

Healthy Bites wishes to serve the orders as early as possible. The individual items in any order are served as and when ready; however, the order is considered to be completely served only when all the items of that order are served.

The table below gives the orders of three clients and the times at which they placed their orders:

Client no.	Time	Order
1	10:00	1 burger, 3 portions of fries, 1 order of ice cream
2	10:05	2 portions of fries. 1 order of ice cream
3	10:07	1 burger. 1 portion of fries

Question 19

Assume that only one client's order can be processed at any given point of time. So, Anish or Bani cannot start preparing a new order while a previous order is being prepared.

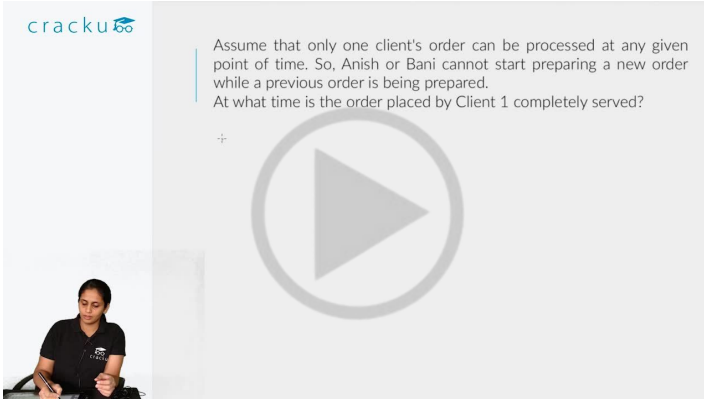
At what time is the order placed by Client 3 completely served?

- A 10:35
- B 10:22
- C 10:25
- D 10:17

cracku

Assume that only one client's order can be processed at any given point of time. So, Anish or Bani cannot start preparing a new order while a previous order is being prepared.
At what time is the order placed by Client 1 completely served?

+



VIDEO SOLUTION

Question 20

Suppose the employees are allowed to process multiple orders at a time, but the preference would be to finish orders of clients who placed their orders earlier.

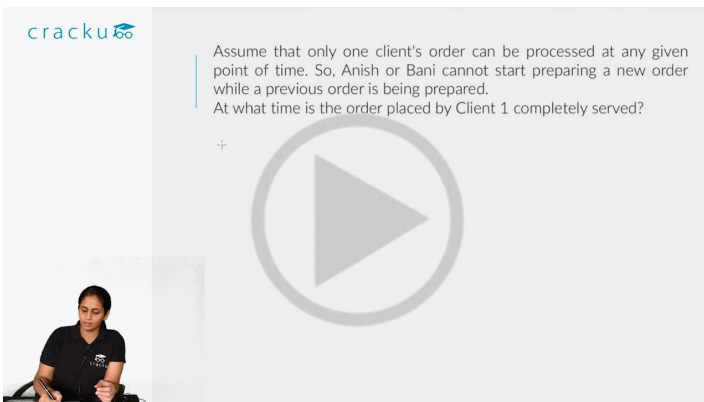
At what time is the order placed by Client 2 completely served?

- A 10:10
- B 10:12
- C 10:15
- D 10:17

cracku

Assume that only one client's order can be processed at any given point of time. So, Anish or Bani cannot start preparing a new order while a previous order is being prepared.
At what time is the order placed by Client 1 completely served?

+



VIDEO SOLUTION

Instructions

Three doctors, Dr. Ben, Dr. Kane and Dr. Wayne visit a particular clinic Monday to Saturday to see patients. Dr. Ben sees each patient for 10 minutes and charges Rs. 100/-. Dr. Kane sees each patient for 15 minutes and charges Rs. 200/-, while Dr. Wayne sees each patient for 25 minutes and charges Rs. 300/-. The clinic has three rooms numbered 1, 2 and 3 which are assigned to the three doctors as per the following table.

Room No.	Monday & Tuesday	Wednesday & Thursday	Friday & Saturday
1	Ben	Wayne	Kane
2	Kane	Ben	Wayne
3	Wayne	Kane	Ben

The clinic is open from 9 a.m. to 11.30 a.m. every Monday to Saturday.

On arrival each patient is handed a numbered token indicating their position in the queue, starting with token number 1 every day. As soon as any doctor becomes free, the next patient in the queue enters that emptied room for consultation. If at any time, more than one room is free then the waiting patient enters the room with the smallest number. For example, if the next two patients in the queue have token numbers 7 and 8 and if rooms numbered 1 and 3 are free, then patient with token number 7 enters room number 1 and patient with token number 8 enters room number 3.

Question 21

The queue is never empty on one particular Saturday. Which of the three doctors would earn the maximum amount in consultation charges on that day?

- A Dr. Wayne
- B Dr. Kane
- C Dr. Ben
- D Both Dr. Wayne and Dr. Kane

cracku

while Dr. Wayne sees each patient for 25 minutes and charges Rs. 300/-. The clinic has three rooms numbered 1, 2 and 3 which are assigned to the three doctors as per the following table.

Room No.	Monday & Tuesday	Wednesday & Thursday	Friday & Saturday
1	Ben	Wayne	Kane
2	Kane	Ben	Wayne
3	Wayne	Kane	Ben

The clinic is open from 9 a.m. to 11.30 a.m. every Monday to Saturday.

On arrival each patient is handed a numbered token indicating their position in the queue, starting with token number 1 every day. As soon as any doctor becomes free, the next patient in the queue enters that emptied room for consultation. If at any time, more than one room is free then the waiting patient enters the room with the smallest number. For example, if the next two patients in the queue have token numbers 7 and 8 and if rooms numbered 1 and 3 are free, then patient with token number 7 enters room number 1 and patient with token number 8 enters room number 3.

Question (1)
What is the maximum number of patients that the clinic can cater to on any single day?

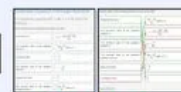
9-11:30
b = 150 - 15 = 135



VIDEO SOLUTION



CAT FORMULAS REVISION



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Question 22

Mr. Singh visited the clinic on Monday, Wednesday, and Friday of a particular week, arriving at 8:50 a.m. on each of the three days. His token number was 13 on all three days. On which day was he at the clinic for the maximum duration?

- A Same duration on all three days
- B Friday
- C Monday
- D Wednesday

Cracku

While Dr. Wayne sees each patient for 25 minutes and charges Rs. 300/-. The clinic has three rooms numbered 1, 2 and 3 which are assigned to the three doctors as per the following table.

Room No.	Monday & Tuesday	Wednesday & Thursday	Friday & Saturday
1	Ben	Wayne	Kane
2	Kane	Ben	Wayne
3	Wayne	Kane	Ben

The clinic is open from 9 a.m. to 11.30 a.m. every Monday to Saturday.

On arrival each patient is handed a numbered token indicating their position in the queue, starting with token number 1 every day. As soon as any doctor becomes free, the next patient in the queue enters that emptied room for consultation. If at any time, more than one room is free then the waiting patient enters the room with the smallest number. For example, if the next two patients in the queue have token numbers 7 and 8 and if rooms numbered 1 and 3 are free, then patient with token number 7 enters room number 1 and patient with token number 8 enters room number 3.

Question (1)

What is the maximum number of patients that the clinic can cater to on any single day?

Handwritten notes: 9-11:30, 25 min, $b = 150 = 15$

VIDEO SOLUTION

Question 23

On a slow Thursday, only two patients are waiting at 9 a.m. After that two patients keep arriving at exact 15-minute intervals starting at 9:15 a.m. – i.e. at 9:15 a.m., 9:30 a.m., 9:45 a.m. etc. Then the total duration in minutes when all three doctors are simultaneously free is

- A 30
- B 10
- C 15
- D 0

Cracku

While Dr. Wayne sees each patient for 25 minutes and charges Rs. 300/-. The clinic has three rooms numbered 1, 2 and 3 which are assigned to the three doctors as per the following table.

Room No.	Monday & Tuesday	Wednesday & Thursday	Friday & Saturday
1	Ben	Wayne	Kane
2	Kane	Ben	Wayne
3	Wayne	Kane	Ben

The clinic is open from 9 a.m. to 11.30 a.m. every Monday to Saturday.

On arrival each patient is handed a numbered token indicating their position in the queue, starting with token number 1 every day. As soon as any doctor becomes free, the next patient in the queue enters that emptied room for consultation. If at any time, more than one room is free then the waiting patient enters the room with the smallest number. For example, if the next two patients in the queue have token numbers 7 and 8 and if rooms numbered 1 and 3 are free, then patient with token number 7 enters room number 1 and patient with token number 8 enters room number 3.

Question (1)

What is the maximum number of patients that the clinic can cater to on any single day?

Handwritten notes: 9-11:30, 25 min, $b = 150 = 15$

VIDEO SOLUTION

Instructions

The Humanities department of a college is planning to organize eight seminars, one for each of the eight doctoral students - A, B, C, D, E, F, G and H. Four of them are from Economics, three from Sociology and one from Anthropology department. Each student is guided by one among P, Q, R, S and T. Two students are guided by each of P, R and T, while one student is guided by each of Q and S. Each student is guided by a guide belonging to their department. Each seminar is to be scheduled in one of four consecutive 30-minute slots starting at 9:00 am, 9:30 am, 10:00 am and 10:30 am on the same day. More than one seminars can be scheduled in a slot, provided the guide is free. Only three rooms are available and hence at the most three seminars can be scheduled in a slot. Students who are guided by the same guide must be scheduled in consecutive slots.

The following additional facts are also known.

1. Seminars by students from Economics are scheduled in each of the four slots.
2. A's is the only seminar that is scheduled at 10:00 am. A is guided by R.
3. F is an Anthropology student whose seminar is scheduled at 10:30 am.
4. The seminar of a Sociology student is scheduled at 9:00 am.
5. B and G are both Sociology students, whose seminars are scheduled in the same slot. The seminar of an Economics student, who is guided by T, is also scheduled in the same slot.
6. P, who is guiding both B and C, has students scheduled in the first two slots.
7. A and G are scheduled in two consecutive slots.

Question 24

Which of the following statements is necessarily true?

- A Q is guiding G.
- B H is an Economics student.
- C S is guiding F.
- D B is scheduled in the first slot.



B -
 C -
 D -
 E -
 F -
 G -
 H -

(2) P -
 (1) Q -
 (2) R -
 (1) S -
 (2) T -

9:00	
9:30	
10:00	
10:30	

10:30 am on the same day. More than one seminars can be scheduled in a slot, provided the guide is free. Only three rooms are available and hence at the most three seminars can be scheduled in a slot. Students who are guided by the same guide must be scheduled in consecutive slots.

The following additional facts are also known.

1. Seminars by students from Economics are scheduled in each of the four slots.
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4. The seminar of a Sociology student is scheduled at 9:00 am.
5. B and G are both Sociology students, whose seminars are scheduled in the same slot. The seminar of an Economics student, who is guided by T, is also scheduled in the same slot.
6. P, who is guiding both B and C, has students scheduled in the first two slots.
7. A and G are scheduled in two consecutive slots.

VIDEO SOLUTION

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Question 25

If D is scheduled in a slot later than Q's, then which of the following two statement(s) is(are) true?

- (i) E and H are guided by T.
- (ii) G is guided by Q.

- A Only (ii)
- B Neither (i) nor (ii)
- C Only (i)
- D Both (i) and (ii)

Cracku

(2) P -
(1) Q -
(2) R -
(1) S -
(2) T -

9:00	
9:30	
10:00	
10:30	

10:30 am on the same day. More than one seminars can be scheduled in a slot, provided the guide is free. Only three rooms are available and hence at the most three seminars can be scheduled in a slot. Students who are guided by the same guide must be scheduled in consecutive slots.

The following additional facts are also known.

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2. A's is the only seminar that is scheduled at 10:00 am. A is guided by R.
3. F is an Anthropology student whose seminar is scheduled at 10:30 am.
4. The seminar of a Biology student is scheduled at 9:00 am.
5. B and G are both Sociology students, whose seminars are scheduled in the same slot. The seminar of an Economics student, who is guided by T, is scheduled in the same slot.
6. The seminar of a student who is guided by both R and C, has students scheduled in the first two slots.
7. B and G are scheduled in two consecutive slots.

VIDEO SOLUTION

Question 26

If E and Q are both scheduled in the same slot, then which of the following statements BEST describes the relationship between D, H, and T?

- A Exactly one of D and H is guided by T.
- B Both D and H are guided by T.
- C At least one of D and H is guided by T.
- D Neither D nor H is guided by T.

Cracku

(2) P -
(1) Q -
(2) R -
(1) S -
(2) T -

9:00	
9:30	
10:00	
10:30	

10:30 am on the same day. More than one seminars can be scheduled in a slot, provided the guide is free. Only three rooms are available and hence at the most three seminars can be scheduled in a slot. Students who are guided by the same guide must be scheduled in consecutive slots.

The following additional facts are also known.

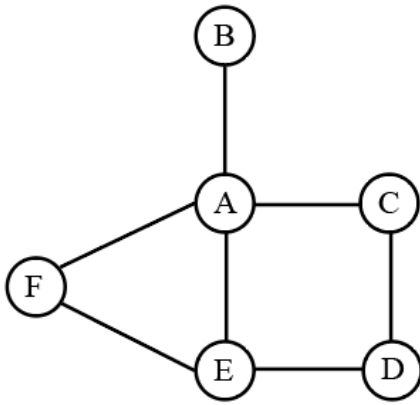
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3. F is an Anthropology student whose seminar is scheduled at 10:30 am.
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5. B and G are both Sociology students, whose seminars are scheduled in the same slot. The seminar of an Economics student, who is guided by T, is scheduled in the same slot.
6. The seminar of a student who is guided by both R and C, has students scheduled in the first two slots.
7. B and G are scheduled in two consecutive slots.

VIDEO SOLUTION

Instructions

Comprehension:

A, B, C, D, E and F are the six police stations in an area, which are connected by streets as shown below. Four teams - Team 1, Team 2, Team 3 and Team 4 - patrol these streets continuously between 09:00 hrs. and 12:00 hrs. each day.



The teams need 30 minutes to cross a street connecting one police station to another. All four teams start from Station A at 09:00 hrs. and must return to Station A by 12:00 hrs. They can also pass via Station A at any point on their journeys.

The following facts are known.

1. None of the streets has more than one team traveling along it in any direction at any point in time.
2. Teams 2 and 3 are the only ones in stations E and D respectively at 10:00 hrs.
3. Teams 1 and 3 are the only ones in station E at 10:30 hrs.
4. Teams 1 and 4 are the only ones in stations B and E respectively at 11:30 hrs.
5. Team 1 and Team 4 are the only teams that patrol the street connecting stations A and E.
6. Team 4 never passes through Stations B, D or F.

Question 27

Which one among the following stations is visited the largest number of times?

- A Station C
- B Station E
- C Station D
- D Station F

	9	9:30	10	10:30	11	11:30	12
Team 1	A						A
Team 2	A		E				A
Team 3	A		D				A
Team 4	A						A

The teams need 30 minutes to cross a street connecting one police station to another. All four teams start from Station A at 09:00 hrs. and must return to Station A by 12:00 hrs. They can also pass via Station A at any point on their journeys.

The following facts are known:

1. None of the streets has more than one team traveling along it in any direction at any point in time.
2. Teams 2 and 3 are the only ones in stations E and D respectively at 10:00 hrs.
3. Teams 1 and 3 are the only ones in station E at 10:30 hrs.
4. Teams 1 and 4 are the only ones in stations B and E respectively at 11:30 hrs.
5. Team 1 and Team 4 are the only teams that patrol the street connecting stations A and E.
6. Team 4 never passes through Stations B, D or F.

VIDEO SOLUTION

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CAT 2024 SYLLABUS



Instructions

Seven candidates, Akil, Balaram, Chitra, Divya, Erina, Fatima, and Ganeshan, were invited to interview for a position. Candidates were required to reach the venue before 8 am. Immediately upon arrival, they were sent to one of three interview rooms: 101, 102, and 103. The following venue log shows the arrival times for these candidates. Some of the names have not been recorded in the log and have been marked as '?'.

Time	7:10 AM	7:15 AM	7:25 AM	7:30 AM	7:40 AM	7:45 AM
Person	Akil, ?	?	?	Chitra	Fatima	?

Additionally here are some statements from the candidates:

Balaram: I was the third person to enter Room 101.

Chitra: I was the last person to enter the room I was allotted to.

Erina: I was the only person in the room I was allotted to.

Fatima: Three people including Akhil were already in the room that I was allotted to when I entered it.

Ganeshan : I was one among the two candidates allotted to Room 102.

Question 28

When did Erina reach the venue?

- A 7:45 am
- B 7:25 am
- C 7:15 am
- D 7:10 am

[VIDEO SOLUTION](#)

Question 29

Who else was in Room 102 when Ganeshan entered?

- A Akil
- B Divya
- C Chitra
- D No one

cracku

Seven candidates, Akh, Balaram, Chitra, Divya, Erina, Fatima, and Ganeshan, were invited to interview for a position. Candidates were required to reach the venue before 8 am. Immediately upon arrival, they were sent to one of three interview rooms: 101, 102, and 103. The following venue log shows the arrival times for these candidates. Some of the names have not been recorded in the log and have been marked as "?".

Time	17:10 AM	7:15 AM	7:20 AM	7:30 AM	7:40 AM	7:45 AM
Person	Akh	?	?	?	Chitra	?

Additionally here are some statements from the candidates:

- Balaram: I was the third person to enter Room 101.
- Chitra: I was the last person to enter the room I was allotted to.
- Erina: I was the only person in the room I was allotted to.
- Fatima: Three people including Akh were already in the room that I was allotted to when I entered it.
- Ganeshan: I was one among the two candidates allotted to Room 102.

Time

A - 7:10

B -

C - 7:30

D -

E -

G -

101

102

103

x101

40

VIDEO SOLUTION

Question 30

What best can be said about the room to which Divya was allotted?

- A Definitely Room 101
- B Definitely Room 103
- C Definitely Room 102
- D Either Room 101 or Room 102

cracku

Seven candidates, Akh, Balaram, Chitra, Divya, Erina, Fatima, and Ganeshan, were invited to interview for a position. Candidates were required to reach the venue before 8 am. Immediately upon arrival, they were sent to one of three interview rooms: 101, 102, and 103. The following venue log shows the arrival times for these candidates. Some of the names have not been recorded in the log and have been marked as "?".

Time	17:10 AM	7:15 AM	7:20 AM	7:30 AM	7:40 AM	7:45 AM
Person	Akh	?	?	?	Chitra	?

Additionally here are some statements from the candidates:

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- Erina: I was the only person in the room I was allotted to.
- Fatima: Three people including Akh were already in the room that I was allotted to when I entered it.
- Ganeshan: I was one among the two candidates allotted to Room 102.

Time

A - 7:10

B -

C - 7:30

D -

E -

G -

101

102

103

x101

40

VIDEO SOLUTION

[CAT Formulas PDF \[Download Now\]](#)

Instructions

A journal plans to publish 18 research papers, written by eight authors (A, B, C, D, E, F, G, and H) in four issues of the journal scheduled in January, April, July and October. Each of the research papers was written by exactly one of the eight authors. Five papers were scheduled in each of the first two issues, while four were scheduled in each of the last two issues. Every author wrote at least one paper and at most three papers. The total number of papers written by A, D, G and H was double the total number of papers written by the other four authors. Four of the authors were from India and two each were from Japan and China. Each author belonged to exactly one of the three areas — Manufacturing, Automation, and Logistics. Four of the authors were from the Logistics area and two were from the Automation area. As per the journal policy, none of the authors could have more than one paper in any issue of the journal.

The following facts are also known.

1. F, an Indian author from the Logistics area, wrote only one paper. It was scheduled in the October issue.
2. A was from the Automation area and did not have a paper scheduled in the October issue.
3. None of the Indian authors were from the Manufacturing area and none of the Japanese or Chinese authors were from the Automation area.
4. A and H were from different countries, but had their papers scheduled in exactly the same issues.
5. C and E, both Chinese authors from different areas, had the same number of papers scheduled. Further, E had papers scheduled in consecutive issues of the journal but C did not.
6. B, from the Logistics area, had a paper scheduled in the April issue of the journal.
7. B and G belonged to the same country. None of their papers were scheduled in the same issue of the journal.
8. D, a Japanese author from the Manufacturing area, did not have a paper scheduled in the July issue.
9. C and H belonged to different areas.

Question 31

Which of the following statements is FALSE?

- A There were exactly two papers by authors from Manufacturing area in the January issue.
- B There was exactly one paper by an author from Manufacturing area in the April issue.
- C There was exactly one paper by an author from Logistics area in the October issue.
- D There were exactly two papers by authors from Manufacturing area in the July issue.

cracku

$A+D+G+H = 2(B+E+F)$

$2x + 11 = 2(6 + 4 + 4)$

$2x + 11 = 26$

$2x = 15$

$x = 7.5$

Jan - 5
April - 5
July - 4
Oct - 4

$x + 2x = 18$

$3x = 18$

$x = 6$

A journal plans to publish 18 research papers, written by eight authors (A, B, C, D, E, F, G, and H) in four issues of the journal scheduled in January, April, July and October. Each of the research papers was written by exactly one of the eight authors. Five papers were scheduled in each of the first two issues, while four were scheduled in each of the last two issues. Every author wrote at least one paper and at most three papers. The total number of papers written by A, D, G and H was double the total number of papers written by the other four authors. Four of the authors were from India and two were from Japan and China. Each author belonged to exactly one of the three areas: Manufacturing, Automation, and Logistics. Four of the authors were from the Logistics area and two were from the Automation area. As per the journal policy, none of the authors could have more than one paper in any issue of the journal.

The following facts are also known:

1. F, an Indian author from the Logistics area, wrote only one paper. It was scheduled in the October issue.
2. A was from the Automation area and did not have a paper scheduled in the October issue.
3. None of the Indian authors were from the Manufacturing area and none of the Japanese or Chinese authors were from the Automation area.
4. A and H were from different countries, but had their papers scheduled in exactly the same issues.
5. C and E, both Chinese authors from different areas, had the same number of papers scheduled. Further, E had papers scheduled in consecutive issues of the journal but C did not.
6. B, from the Logistics area, had a paper scheduled in the April issue of the journal.
7. B and G belonged to the same country. None of their papers were scheduled in the same issue of the journal.
8. D, a Japanese author from the Manufacturing area, did not have a paper scheduled in the July issue.
9. C and H belonged to different areas.

VIDEO SOLUTION

Question 32

How many papers were written by Indian authors?

cracku

$A+D+G+H = 2(B+E+F)$

$2x + 11 = 2(6 + 4 + 4)$

$2x + 11 = 26$

$2x = 15$

$x = 7.5$

Jan - 5
April - 5
July - 4
Oct - 4

$x + 2x = 18$

$3x = 18$

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VIDEO SOLUTION

Instructions

Three doctors, Dr. Ben, Dr. Kane and Dr. Wayne visit a particular clinic Monday to Saturday to see patients. Dr. Ben sees each patient for 10 minutes and charges Rs. 100/-. Dr. Kane sees each patient for 15 minutes and charges Rs. 200/-, while Dr. Wayne sees each patient for 25 minutes and charges Rs. 300/-. The clinic has three rooms numbered 1, 2 and 3 which are assigned to the three doctors as per the following table.

Room No.	Monday & Tuesday	Wednesday & Thursday	Friday & Saturday
1	Ben	Wayne	Kane
2	Kane	Ben	Wayne
3	Wayne	Kane	Ben

The clinic is open from 9 a.m. to 11.30 a.m. every Monday to Saturday.

On arrival each patient is handed a numbered token indicating their position in the queue, starting with token number 1 every day. As soon as any doctor becomes free, the next patient in the queue enters that emptied room for consultation. If at any time, more than one room is free then the waiting patient enters the room with the smallest number. For example, if the next two patients in the queue have token numbers 7 and 8 and if rooms numbered 1 and 3 are free, then patient with token number 7 enters room number 1 and patient with token number 8 enters room number 3.

Question 33

What is the maximum number of patients that the clinic can cater to on any single day?

- A 12
- B 30
- C 31
- D 15

Question (1)

What is the maximum number of patients that the clinic can cater to on any single day?

9 - 11:30
2.5 hours
2.5 hours = 150 minutes

VIDEO SOLUTION



CAT FORMULAS BOOK
BY 100%ILER



Instructions

Seven candidates, Akil, Balaram, Chitra, Divya, Erina, Fatima, and Ganeshan, were invited to interview for a position. Candidates were required to reach the venue before 8 am. Immediately upon arrival, they were sent to one of three interview rooms: 101, 102, and 103. The following venue log shows the arrival times for these candidates. Some of the names have not been recorded in the log and have been marked as '?'.
101
102
103

Time	7:10 AM	7:15 AM	7:25 AM	7:30 AM	7:40 AM	7:45 AM
Person	Akil, ?	?	?	Chitra	Fatima	?

Additionally here are some statements from the candidates:

Balaram: I was the third person to enter Room 101.

Chitra: I was the last person to enter the room I was allotted to.

Erina: I was the only person in the room I was allotted to.

Fatima: Three people including Akhil were already in the room that I was allotted to when I entered it.

Ganeshan : I was one among the two candidates allotted to Room 102.

Question 34

If Ganeshan entered the venue before Divya, when did Balaram enter the venue?

- A 7:25 am
- B 7:10 am
- C 7:15 am
- D 7:45 am